

Kevin Ato Ocran

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PROFESSIONAL SUMMARY

Data Science and Analytics professional with hands-on experience architecting and deploying production-grade machine learning systems on GCP. Demonstrated ability to lead cross-functional ML and Cloud Engineering teams, design end-to-end MLOps workflows, and translate raw, unlabelled datasets into actionable intelligence. Core competencies span unsupervised anomaly detection, predictive modelling, cloud infrastructure automation, and the responsible integration of Generative AI tools. Actively seeking opportunities to apply these skills in impactful, real-world environments.

TECHNICAL SKILLS

Languages: Python, SQL, Java, Bash

Data & ML: pandas, NumPy, scikit-learn, XGBoost, MLflow, Matplotlib, Seaborn, BeautifulSoup, Jupyter Notebooks

Cloud & DevOps: GCP (Cloud Run, Cloud Storage, Artifact Registry, Cloud Scheduler, Cloud Monitoring), Docker, Git & GitHub

Databases & Tools: PostgreSQL, Supabase, Excel, PowerPoint

Generative AI & LLMs: Prompt Engineering, Meta-Prompt Engineering, LLM Application Integration (Claude, OpenAI API)

PROFESSIONAL EXPERIENCE

Cloud & ML Team Lead (Intern) · Thrive Africa

2025 · 3 months

- Spearheaded operations for both the Machine Learning and Cloud Engineering teams; orchestrated cross-functional collaboration to deliver a unified enterprise project, managing version control (GitHub) and Agile sprint boards (Trello).
- Established weekly operational sprints, delegated technical milestones, and conducted structured one-on-one mentorship sessions to optimise team performance and ensure timely deliverables.
- Engineered an Isolation Forest anomaly detection model on the UCI Household Power Consumption dataset (~2.07M rows, 4 years of minute-level readings), achieving a competition-winning validation False Positive Rate of 1.4809%.
- Deployed the predictive model as a FastAPI microservice on GCP Cloud Run; integrated MLflow for full experiment tracking and model registry; implemented automated weekly retraining pipelines via Cloud Scheduler.
- Architected end-to-end GCP infrastructure comprising Artifact Registry, Cloud Storage, an MLflow server, and a production model hot-swap endpoint; designed Cloud Monitoring dashboards with 4 critical KPI widgets.
- Leveraged advanced Generative AI tools to rapidly prototype and deploy a highly responsive React/Next.js web front-end for the live anomaly detection platform.

TECHNICAL PROJECTS

Energy Consumption Anomaly Detector — *Python · scikit-learn · FastAPI · Docker · GCP · MLflow · Next.js · Supabase*
github.com/cloud-ml-internship-2026/group-d1 | [Live Demo](#)

- Architected an unsupervised Isolation Forest anomaly detection pipeline on the UCI Household Power Consumption dataset (2.07M rows, Dec 2006 – Nov 2010), engineering 16 advanced features including cyclical time encodings (hour_sin/cos), rolling deviation metrics, and lag features.
- Implemented Infrastructure as Code (IaC) principles via deployment scripts (infra/setup.sh) and configuration files (service.yaml) to provision Cloud Run services, Cloud Monitoring dashboards, and Cloud Scheduler jobs.
- Achieved a validation FPR of 1.4809% (competition metric); tracked 15 model versions in MLflow with a @champion alias system enabling zero-downtime hot-swaps; deployed with automated Sunday-midnight retraining and a p99 latency target < 2s.

Open-Source Project Health Analysis — Python · pandas · Matplotlib · Seaborn · GitHub API

github.com/KevinOcran/openSourceHealthAnalysis

- Formulated an ETL pipeline extracting 100 issues and commits each from 5 major repositories (Pandas, NumPy, scikit-learn, Airflow, MLflow) via the GitHub REST API, with rate-limit handling and nested JSON normalisation.
- Designed and calculated advanced data metrics, including Issue Lifecycle Latency and Contributor Distribution Indexes, to evaluate repository sustainability; produced 6 publication-quality visualisations and 3 actionable maintenance recommendations.

Online Retail Data Pipeline & Star Schema — Python · pandas · PostgreSQL · SQLAlchemy · Excel

github.com/KevinOcran/online_retail

- Cleansed, normalised, and partitioned 541,910 raw e-commerce transactional records, elevating data quality from ~70% to 95%+ clean records; designed and implemented a Star Schema in PostgreSQL with fact and dimension tables.
- Executed advanced SQL validation scripts, including complex multi-table joins, to isolate Net Revenue by Country and audit data pipeline integrity.

Titanic Survival Prediction — Python · scikit-learn · XGBoost · pandas · Matplotlib

github.com/KevinOcran/titanic

- Developed an end-to-end predictive classification pipeline featuring EDA, advanced feature engineering (log-transforms, scaling, categorical encoding), and missing-value imputation; benchmarked Logistic Regression, Random Forest, and XGBoost.
- Achieved ~83% test accuracy (vs. 62% baseline) with a tuned Random Forest; evaluated with confusion matrix, ROC/AUC curves, and feature importance plots.

EDUCATION

BSc. Data Science and Analytics · Ghana Communication Technology University (GCTU) 2024 – Present

Expected Graduation: June 2028 · Accra, Ghana

CERTIFICATIONS

- **Machine Learning and AI** — Thrive Africa
- **Data Science and Analytics** — Thrive Africa
- **Generative AI and Advanced Prompt Engineering** — Thrive Africa
- **AWS Cloud Computing Essentials** — Thrive Africa
- **Prompt Engineering for Developers** — DeepLearning.AI & OpenAI

VOLUNTEER EXPERIENCE

Mathematics & IT Teacher · Ricin Foundation School · Assemblies of God Ghana 2024 – 2025 · 6 months

- Delivered Mathematics and Information Technology lessons to learners from Primary 5 through JHS 3 in an under-resourced community school.
- Designed engaging lesson plans and practical exercises to cultivate foundational numeracy and digital literacy skills.